Necessity Entrepreneurs

Occupational Choice, Misallocation, and Unemployment Insurance

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2025

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Roadmap

Motivation

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- **Big picture:** Entrepreneurship drives innovation and growth

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Key Distinction
Opportunity entrepreneurs choose entrepreneurship regardless of employment status

Necessity entrepreneurs would *not* start a business if they had a job

Research Question:

How does unemployment risk shape
the composition of entrepreneurship?

What we do:

- Build a span-of-control model with search frictions
- Endogenous occupational choice $\rightarrow$ two types of entrepreneurs emerge

Preview of findings:

- Unemployment insurance reduces necessity entrepreneurship
- Consumption floors have limited effect when UI is present

Model

Economic Environment

Agents differ in human capital h (education: P, S, or T)

h determines:

- Managerial ability x (Pareto distributed)
- Labour productivity z
- Job separation rate ρ

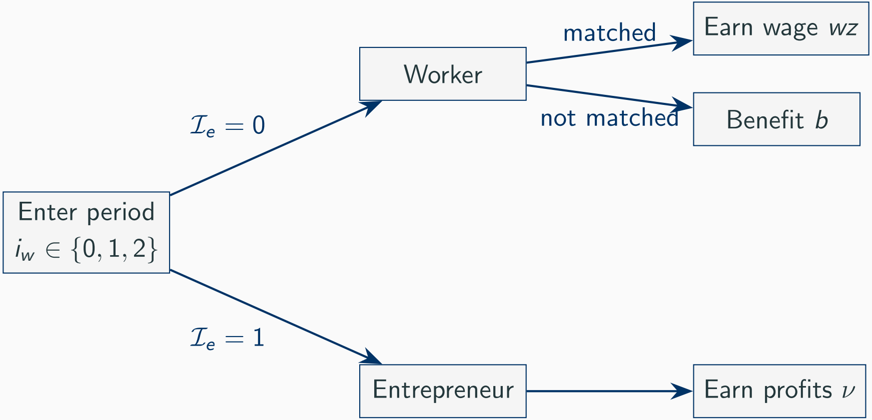
Production (Lucas span-of-control):

$$y = xk^{\alpha}(zn)^{\gamma} \qquad \alpha, \gamma > 0$$

Labour market: Search and matching frictions

- Employed workers face separation probability ρ
- Unemployed workers match with probability $p(\theta)$

Timing and Occupational Choice



Key insight: Employment status i_w affects the *outside option*, so it affects the entrepreneurship threshold $\hat{h}$

Key Mechanism: Two Types of Entrepreneurs

Let $\hat{h}$ be the critical human capital threshold for entrepreneurship.

Threshold depends on employment status:

$$\hat{h}_{i_w \neq 1} < \hat{h}_{i_w = 1}$$

Unmatched workers have a *lower* threshold $\rightarrow$ easier to push into entrepreneurship

Necessity Entrepreneurs

$$\hat{h}_{i_w \neq 1} < h < \hat{h}_{i_w = 1}$$

Would *not* be entrepreneurs if employed

Opportunity Entrepreneurs

$$h \geq \hat{h}_{i_w = 1} \geq \hat{h}_{i_w \neq 1}$$

Choose entrepreneurship *regardless*

$\rightarrow$ Six critical thresholds total (2 per education level: P, S, T)

Quantitative Analysis

Calibration

Parameter	Value	Description	Source
β	0.94	Discount factor	
σ	2.0	Risk aversion	$u(c) = \frac{c^{1-\sigma}}{1-\sigma}$
α	0.3207	Capital share	
γ	0.4693	Labour share	Frac. of entrepreneurs
$\bar{m}$	0.8	Matching efficiency	Lubik (2013)
μ	$[0.5, 0.7]$	Matching elasticity	Petrongolo & Pissarides
b	$0.40 \cdot w$	UI benefits	Shimmer et al. (2005)
ρ	by h	Separation rate	Cairo & Cajner (2016)
z	by h	Wage premium	Goldin & Katz (2009)

Three education levels (P, S, T) with population shares from Barro & Lee (2013)

Baseline Results: No Unemployment Counterfactual

	Baseline	No Unemployment
Entrepreneur rate (avg)	0.145	0.197
Low education	0.143	0.207
Mid education	0.137	0.184
High education	0.161	0.216
Firm size (avg)	21.9	24.1
Low education	9.7	9.9
Mid education	18.0	19.8
High education	33.4	37.2
Unemp-Ent rate	0.00204	0

Takeaway: Removing unemployment risk *increases* entrepreneurship—all entrants are opportunity-driven, with larger firms

Policy Experiment: Unemployment Insurance

	Baseline (UI= 0.4)	UI= 0.3	UI= 0.6
Entrepreneur rate (avg)	0.145	0.145	0.144
Low education	0.143	0.141	0.135
Mid education	0.137	0.136	0.141
High education	0.161	0.164	0.154
Firm size (avg)	21.9	21.6	21.2
Unemp-Ent rate	0.00204	0.00197	0.00084

Takeaway: Higher UI benefits →

- Necessity entrepreneurship falls (better outside option for unemployed)
- Average firm size declines modestly
- Low-education workers most responsive to UI changes

Further Experiments: Consumption Floor & Credit Frictions

Consumption floor ($\underline{c}$): lump-sum transfer or utility penalty when income falls below threshold

- With UI present: consumption floor has **limited additional effect**
- Without unemployment: consumption floor barely moves entrepreneurship rates

Credit friction wedge (Δ): borrowing cost premium for entrepreneurs

- Tighter credit $\rightarrow$ fewer entrepreneurs, smaller firms
- Interacts with asset holdings and occupational choice

$\rightarrow$ **UI design matters more** than consumption floors for reducing necessity entrepreneurship

Conclusion

Takeaways

1. **Necessity entrepreneurs emerge endogenously** from the interaction of unemployment risk and occupational choice
2. **Unemployment insurance reduces necessity entrepreneurship** by improving the outside option for jobless workers
3. **Policy implication:** Well-designed UI can reduce misallocation of talent without substantially lowering overall entrepreneurship

Thank you!

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Appendix: Household's Problem

$$\mathbf{V}(a,h,i_w) = \max_{\{a',c,p(\theta),\mathcal{I}_e\}} \begin{cases} \iota_{i_w=0}\left[(1-\mathcal{I}_e)\mathbb{E}_{p(\theta)}(\cdot) + \mathcal{I}_e(V_e + \beta\mathbb{E}\mathbf{V}(\cdot))\right] \\ \iota_{i_w=1}\left[(1-\mathcal{I}_e)\mathbb{E}_{\rho}(\cdot) + \mathcal{I}_e(V_e + \beta\mathbb{E}\mathbf{V}(\cdot))\right] \\ \iota_{i_w=2}\left[(1-\mathcal{I}_e)\mathbb{E}_{p(\theta)}(\cdot) + \mathcal{I}_e(V_e + \beta\mathbb{E}\mathbf{V}(\cdot))\right] \end{cases}$$

subject to:

$$c + a' \leq (1 + r - \delta)a + inc$$

where *inc* depends on occupation and employment status.

Appendix: Firm's Problem

Firm's profit maximization:

$$\max_{n,k} \; Xk^{\alpha}n^{\gamma} - (1+\tau\psi)wzn - rk - \kappa\frac{\rho zn}{q(\theta)}$$

Optimal labor demand:

$$n^*(k, x, \tilde{w}) = \left[\frac{\gamma xk}{w(1+\psi\tau) + \frac{\rho\kappa}{q(\theta)}}\right]^{\frac{1}{1-\gamma}}$$

Matching function:

$$p(\theta) = \bar{m}\left(\frac{v}{u}\right)^{1-\mu}, \qquad q(\theta) = \bar{m}\left(\frac{v}{u}\right)^{-\mu}$$